Architectural Critique: 6-Agent Formulation Platform

Document Status: Architectural Evaluation & Review (Milestone 2) | **Scope Limit:** Maximum 2 Pages Rendered
Target Proposal: AI-Generated Draft Design ("Proposed Architecture: Multi-Agent Formulation Platform")

1. Executive Summary & Prioritized Scorecard

The draft proposal outlines a distributed 6-agent microservice architecture coordinated by an autonomous orchestrator. While it correctly maps the functional workflow of chemical R&D (literature intake, drafting, costing, scheduling), it suffers from **fundamental safety risks, unvalidated data assumptions, and severe operational over-engineering** for an 8-chemist team. Crucially, the draft delegates deterministic calculations, safety determinations, and physical lab bookings to stochastic LLMs.

Component / Proposal	Priority Rank	Decision	Core Engineering Rationale
Agent 6: Autonomous Lab Booking	**1 (Critical Safety)**	**REJECT**	Uncalibrated LLM confidence cannot book physical lab assets; risks hazard and lab gridlock.
Agent 3: Synthesizing Missing Data	**2 (Critical Regulatory)**	**REJECT**	Inventing missing stability creates serious safety & compliance risks; requires data quarantine.
Agent 2: Fuzzy-Merging RM Names	**3 (Critical Traceability)**	**REJECT**	Conflates distinct chemical grades & suppliers (`RM-101` vs `RM-102` Esterquats); destroys provenance.
Agent 5: LLM Stoichiometry & Costing	**4 (Critical Quality)**	**REJECT**	Stochastic LLMs fail arithmetic; causes mass-balance failure (`T-09` 103%) & UOM errors (`RM-170`).
Agent 4: Trusting Summary 'Pass' Labels	**5 (High Operational)**	**CHANGE**	Promotes unapproved (`T-04`) & contradicted (`T-05` phase separation) trials; hides negative data.
Agent 2: Vector DB for Tabular Data	**6 (Medium Overhead)**	**CHANGE**	Vector embeddings degrade numerical filtering on tiny tabular sets (28 raw materials, 12 trials).
6 Microservices for 8 Chemists	**7 (Medium Overhead)**	**CHANGE**	Distributed tracing and network latency overhead; replace with a bounded Modular Monolith.
Big-Bang Rollout & Thumbs-Up Loop	**8 (Low Signal)**	**CHANGE**	Coarse feedback fails to catch silent drift; replace with phased rollout & technical telemetry.
Core Assistant & Literature Search	**—**	**KEEP**	Structured prompt synthesis and semantic literature retrieval remain valuable accelerators.

2. Detailed Technical Findings & Mandated Architectural Changes

Finding 1: Autonomous Lab Scheduling on Heuristic Confidence (Agent 6) — REJECT

- **Draft Flaw:** Agent 6 books physical incubator/mixer slots when candidate confidence exceeds 0.8, placing the cancellation burden on chemists. Model confidence has zero correlation with physical bench safety.
- **Workbook Evidence:** Lab slots are scarce and cancellations frequent (`BK-2623` cancelled; `BK-2601` corrupted). Automated booking risks physical gridlock, wasted technician prep, and chemical hazards.
- **Mandated Action: Eliminate autonomous booking.** The assistant outputs a *draft trial plan*. Instrument reservation is permitted solely after explicit, role-based digital sign-off from an authenticated chemist.

Finding 2: Synthesizing Missing Stability and Safety Data (Agent 3) — REJECT

- **Draft Flaw:** Agent 3 infers missing stability and skin-safety properties from parametric LLM knowledge to prevent blocked pipeline runs. Inventing missing stability or safety data creates serious safety, compliance, and traceability risks.
- **Workbook Evidence:** Trial `T-11` has missing pH and ongoing stability. For skin-contact softeners containing known sensitizers (`RM-160` isothiazolinones), unverified or fabricated safety properties jeopardize product safety and audit integrity.
- **Mandated Action: Enforce strict missing-data quarantine.** Missing measurements must remain explicitly unknown and be quarantined for qualified chemist review. LLMs must never invent synthetic safety passes.

Finding 3: Fuzzy-Merging Raw Materials and Synonyms (Agent 2) — REJECT

- **Draft Flaw:** Agent 2 automatically merges similar ingredient names via fuzzy matching to produce "one clean list," erasing critical chemical, grade, and supplier distinctions.
- **Workbook Evidence:** In `RM_Master` , `RM-101` (Supplier A, 90% active @ ₹210/kg) and `RM-102` (Supplier B, alternate @ ₹198/kg) share the synonym `TEA-esterquat` . Fuzzy-merging obscures pricing variance, supplier contracts, and active solids content.
- **Mandated Action: Prohibit silent merging.** Enforce relational primary keys (`RM-xxx`). When unmapped tokens occur (e.g. `TEA-esterquat` in `T-04`), generate an ambiguity warning requiring explicit chemist disambiguation.

Finding 4: LLM-Driven Mass Balance and Costing (Agent 5) — REJECT

- **Draft Flaw:** Agent 5 asks the LLM to calculate formulation cost per kg and rebalance ingredient percentages to 100%. Autoregressive token models cannot perform reliable multidimensional arithmetic.
- **Workbook Evidence:** `T-09` already exhibits human error (103.0% sum). LLMs easily miss packaging unit conversions (calculating `RM-170` at ₹32,000/kg instead of ₹160/kg drum basis) or FX rates.

- **Mandated Action: Transfer all math to a deterministic Python barrier.** Formulation percentages must be validated using `Decimal` fixed-point math (Σ w_i = 100.0% ± 0.05%). Costs must be computed via relational joins with `Price_Feed` records and historical FX rates (₹84.00/USD).

Finding 5: Trusting Summary 'Pass' Labels & Suppressing Negative Evidence (Agent 4) — CHANGE

- **Draft Flaw:** Agent 4 promotes candidate formulations by using historical trials marked `Pass` as positive exemplars, completely ignoring negative or failed experiments.
- **Workbook Evidence:** `T-05` is marked `Pass` in `Trials_TX_Softeners`, but raw incubator logs in `Stability_Log` record phase separation at Week 3. `T-04` is ranked #1 in `Summary_Apr` despite zero chemist sign-off (`approved_by = NULL`) and exceeding target viscosity (0.45 Pa·s = 450 cP > 300 cP).
- **Mandated Action: Implement Bidirectional Retrieval with Log-Level Verification.** Ground truth must be derived from raw measurement tables, not unverified summary fields. The engine must actively retrieve negative/failed trials (`T-05`), annotating failure boundaries to guide chemists away from dead ends.

Finding 6: Microservice Architecture & Vector DB for 8 Chemists — CHANGE

- **Draft Flaw:** Running 6 independent microservices and nightly vector DB ingestion for 8 chemists generating ~15 queries/day introduces severe network latency, deployment overhead, and fuzzy numerical retrieval.
- **Workbook Evidence:** Kestrel's dataset comprises only 28 raw materials and 12 trials. Vector embeddings degrade filtering on precise chemical parameters (solids %, pH ranges).
- **Mandated Action: Adopt a Modular Monolith.** Deploy a single FastAPI service backed by relational SQLite/PostgreSQL with BM25 keyword search, eliminating microservice RPC and container sprawl.

Finding 7: Big-Bang Rollout & Coarse Feedback Loops — CHANGE

- **Draft Flaw:** Launching 6 microservices simultaneously with monthly thumbs-up/down feedback creates high integration risk and fails to detect dangerous recipe drift or silent costing errors.
- **Mandated Action: Execute a Phased 8-Week Rollout.** Deploy ingestion and deterministic validation gates first. Shadow-pilot with 2 senior chemists in Lab 1, instrumenting granular telemetry on every run.

3. Architectural Recommendation Summary

The 6-agent draft treats deterministic chemical and mathematical problems as generative text tasks. By replacing distributed microservices with a **Modular Monolith**, enforcing **deterministic Python validation barriers**, and instituting **mandatory human chemist sign-off**, Kestrel establishes an auditable, cost-effective platform ($0.537/chemist/month baseline) ready for reliable laboratory adoption.